



上海科技大学
ShanghaiTech University

EE115 Analog Circuits 模拟电路

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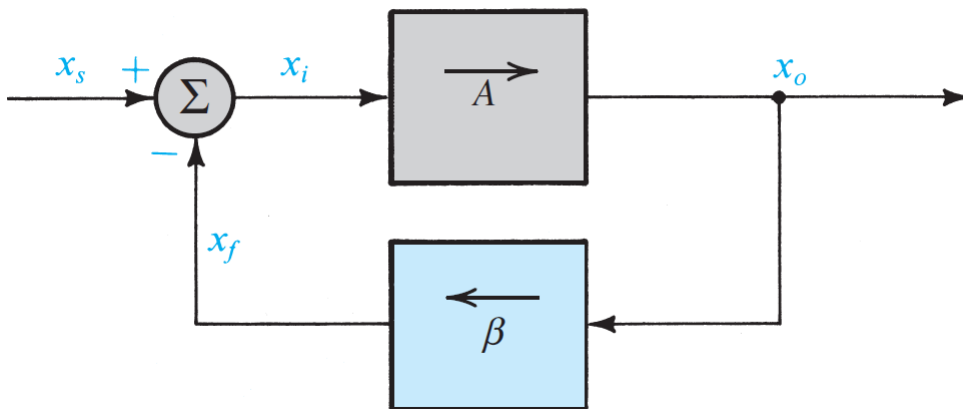
Outline



- Feedback
- SEDTRA/SMITH book page 820-834;

The general feedback structure

- A: open loop gain
- β : feedback factor



$$x_o = Ax_i$$

$$x_f = \beta x_o$$

$$x_i = x_s - x_f$$

$$A_f \equiv \frac{x_o}{x_s}$$

$$A_f = \frac{A}{1 + A\beta}$$

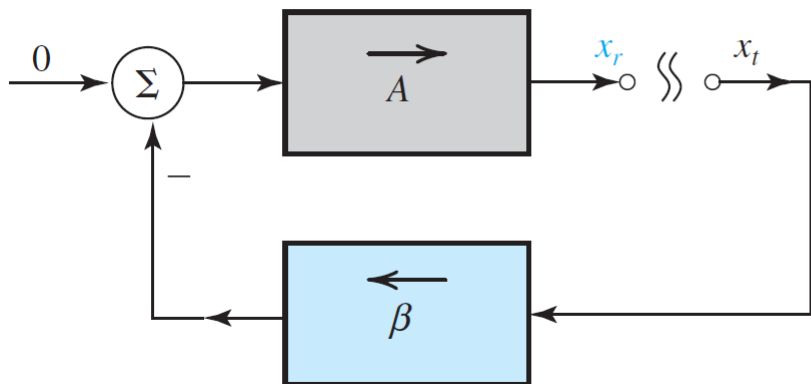
- $A\beta$: loop gain
- $A\beta \gg 1$

$$A_f \simeq \frac{1}{\beta}$$

The loop gain

- $A\beta$: loop gain
- $A\beta \gg 1$
- The sign of $A\beta$ determines the polarity of the feedback
- The magnitude of $A\beta$ determines how close the closed-loop gain A_f is to the ideal value of $1/\beta$.
- The magnitude of $A\beta$ determines the amount of feedback $(1+A\beta)$ the magnitude of the various improvements in amplifier performance

$$A_f \simeq \frac{1}{\beta}$$



$$x_r = -A\beta x_t$$

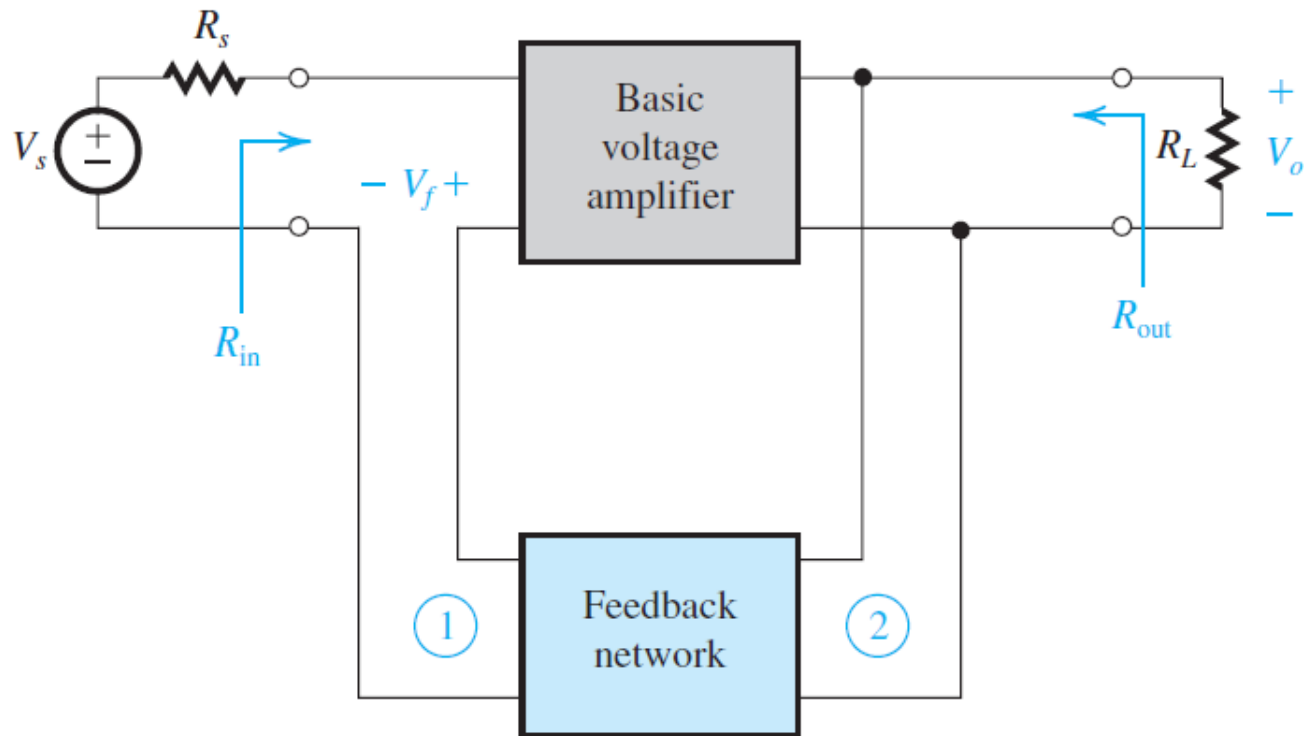
$$A\beta = -\frac{x_r}{x_t}$$

Figure 11.2 Determining the loop gain by breaking the feedback loop at the output of the basic amplifier, applying a test signal x_t , and measuring the returned signal x_r : $A\beta \equiv -x_r/x_t$.

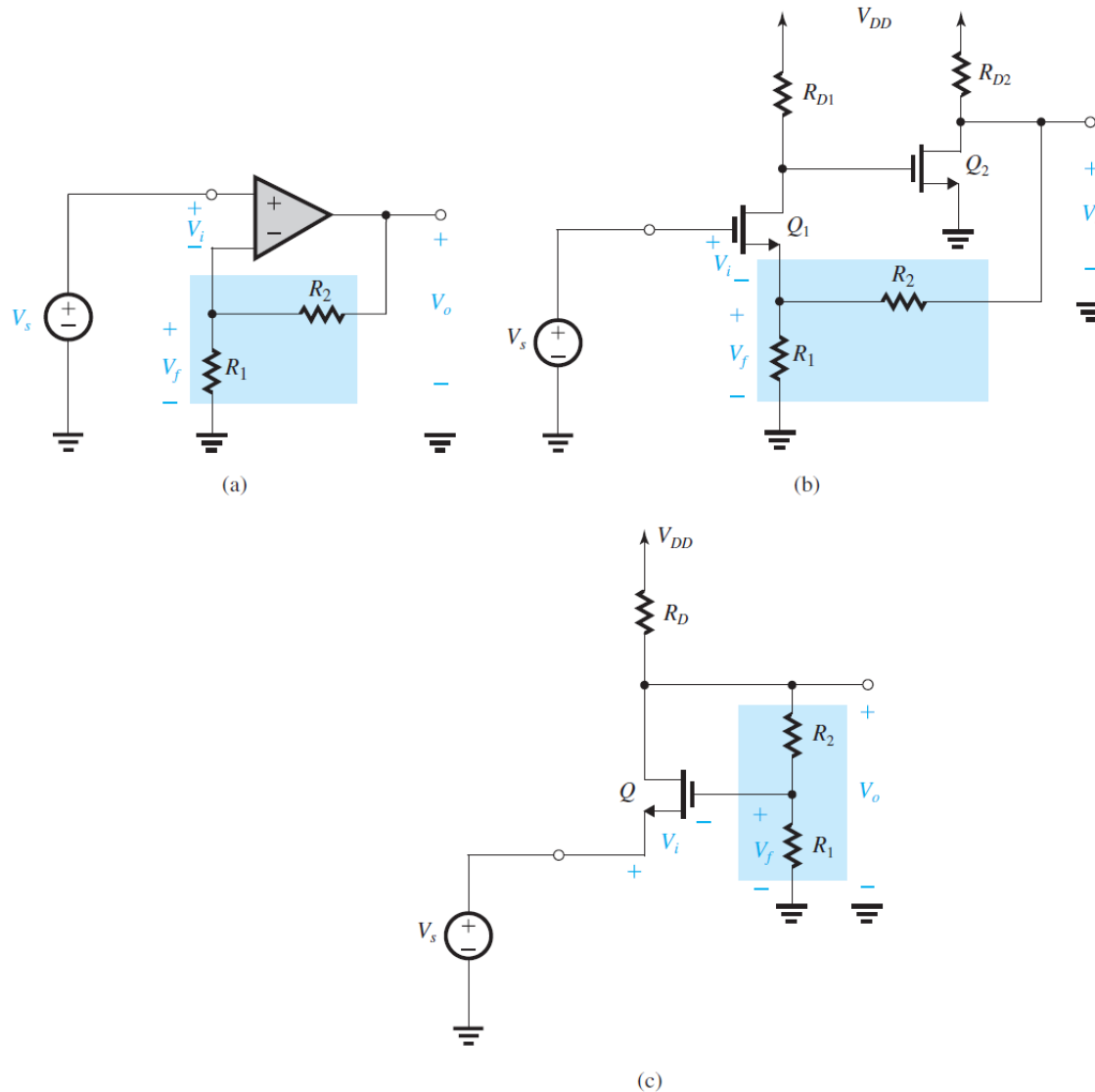
The Feedback Voltage Amplifier



The Series–Shunt Feedback Topology



Examples of Series–Shunt Feedback Amplifiers



Analysis of the Feedback Voltage Amplifier Utilizing the Loop Gain



- Identify the feedback network and determine β
- Determine the ideal value of the closed-loop gain A_f as $1/\beta$
- Determine the loop gain
- Use the value of $A\beta$ together with that of β to determine the open-loop gain A . Then, determine the closed-loop gain A_f :

$$A_f = \frac{A}{1 + A\beta}$$

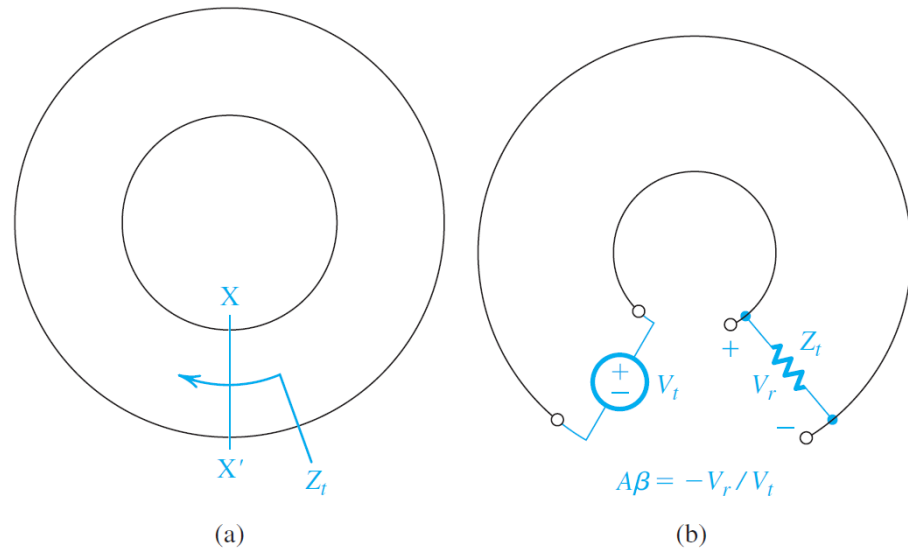
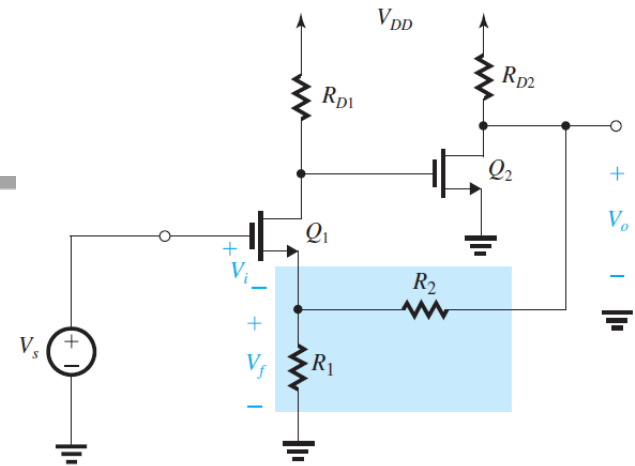


Figure 11.9 Breaking the conceptual feedback loop in (a) to determine the loop gain requires the termination of the loop as shown in (b), to ensure that the loop conditions do not change.

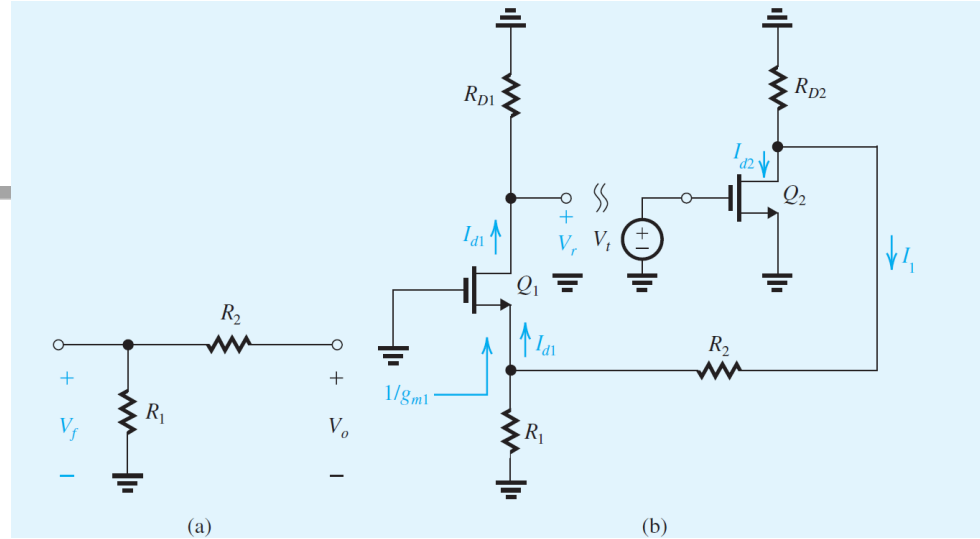
Example



For the series–shunt feedback amplifier of Fig. 11.8(b), neglect the MOSFETs' r_o and

- give the feedback network (β circuit) and an expression for β . Also give an expression for the ideal or upper-bound value of the closed-loop gain A_f .
- find the ratio R_2/R_1 that results in an ideal closed-loop gain of 10 V/V. If $R_1 = 1\text{ k}\Omega$, what value must R_2 have?
- find an expression for the loop gain $A\beta$.
- if $g_{m1} = g_{m2} = 4\text{ mA/V}$ and $R_{D1} = R_{D2} = 10\text{ k}\Omega$, determine the values of $A\beta$, A , and A_f .

Example solution



- (a) The feedback network is highlighted in Fig. 11.8(b) and is redrawn in Fig. 11.10(a). It is a simple resistive voltage divider. Thus,

$$\beta = \frac{V_f}{V_o} = \frac{R_1}{R_1 + R_2}$$

For $A\beta \gg 1$, the closed-loop gain $A_f = V_o/V_s$ is given by

$$A_f \simeq \frac{1}{\beta} = 1 + \frac{R_2}{R_1}$$

This is the ideal or upper-bound value of A_f .

- (b) For A_f to have an ideal value of 10,

$$10 = 1 + \frac{R_2}{R_1}$$

Thus,

$$\frac{R_2}{R_1} = 9$$

For $R_1 = 1 \text{ k}\Omega$, $R_2 = 9 \text{ k}\Omega$.

- (c) To determine the loop gain, we set $V_s = 0$. Examining the feedback loop reveals that it is most convenient to break the loop at the connection between the drain of Q_1 and the gate of Q_2 . This is

Example solution

because of the infinite input impedance at the gate of Q_2 . The resulting circuit is shown in Fig. 11.10(b), for which the analysis to determine $A\beta \equiv -V_r/V_t$ proceeds as follows:

$$\begin{aligned} I_{d2} &= g_{m2} V_t \\ I_1 &= -I_{d2} \frac{R_{D2}}{R_{D2} + R_2 + \left(R_1 \parallel \frac{1}{g_{m1}} \right)} \\ I_{d1} &= I_1 \frac{R_1}{R_1 + \frac{1}{g_{m1}}} \\ V_r &= I_{d1} R_{D1} \end{aligned}$$

Combining these four equations results in

$$A\beta \equiv -\frac{V_r}{V_t} = (g_{m1} R_{D1})(g_{m2} R_{D2}) \frac{1}{1 + g_{m1} R_1} \frac{R_1}{R_{D2} + R_2 + \left(R_1 \parallel \frac{1}{g_{m1}} \right)}$$

(d)

$$\begin{aligned} A\beta &= 4 \times 10 \times 4 \times 10 \times \frac{1}{1 + 4 \times 1} \times \frac{1}{10 + 9 + (1 \parallel \frac{1}{4})} \\ &= 16.67 \end{aligned}$$

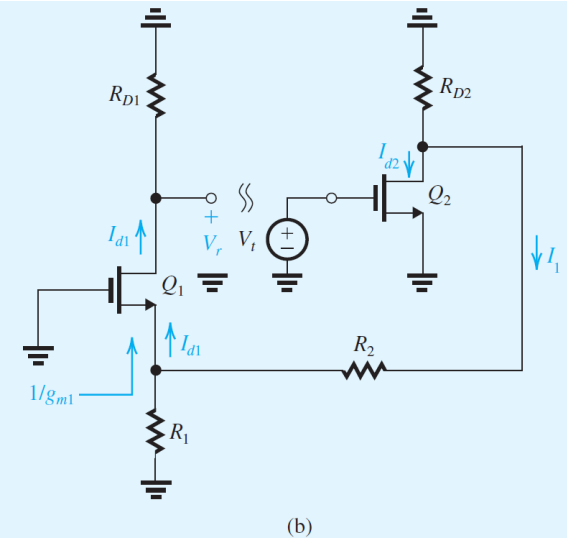
Since

$$\begin{aligned} \beta &= \frac{R_1}{R_1 + R_2} = \frac{1}{1 + 9} = 0.1 \\ A &= \frac{A\beta}{\beta} = \frac{16.67}{0.1} = 166.7 \text{ V/V} \end{aligned}$$

and

$$A_f = \frac{A}{1 + A\beta} = \frac{166.7}{1 + 16.67} = \frac{166.7}{17.67} = 9.43 \text{ V/V}$$

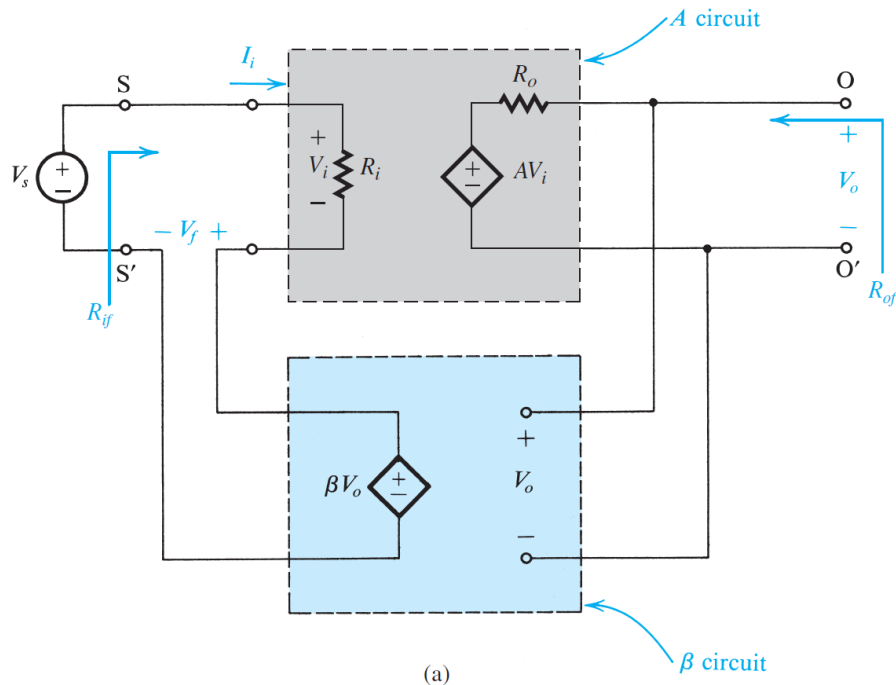
which, given that the loop gain (16.67) is only moderately high, is reasonably close to the ideal value of 10 V/V.



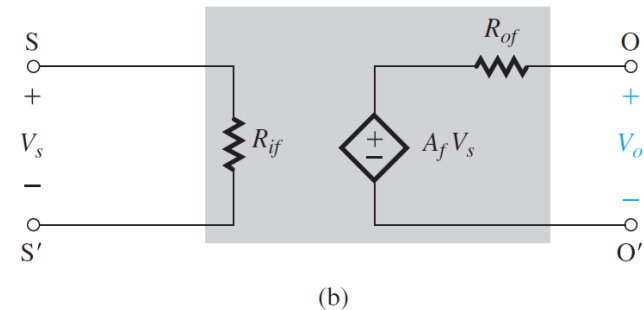
Systematic Analysis of Feedback Voltage Amplifiers



- The ideal structure of the series–shunt feedback amplifier consists of a unilateral open-loop amplifier (the A circuit) and an ideal voltage-sampling, voltage-mixing feedback network (the β circuit). The A circuit has an input resistance R_i , an open-circuit voltage gain A , and an output resistance R_o . It is assumed that the source is ideal with a zero resistance and that there is no load resistance. Also, no loading effect is considered.



$$A_f \equiv \frac{V_o}{V_s} = \frac{A}{1 + A\beta}$$



The series–shunt feedback amplifier: (a) ideal structure; (b) equivalent circuit.

Systematic Analysis of Feedback Voltage Amplifiers

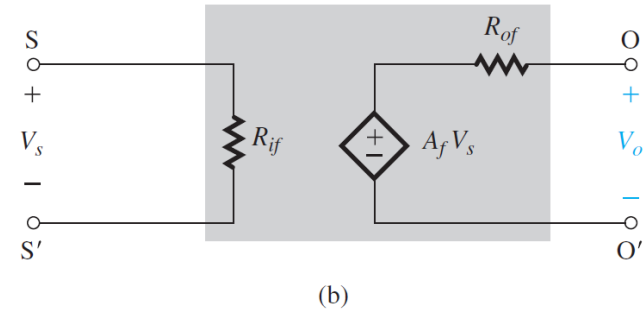
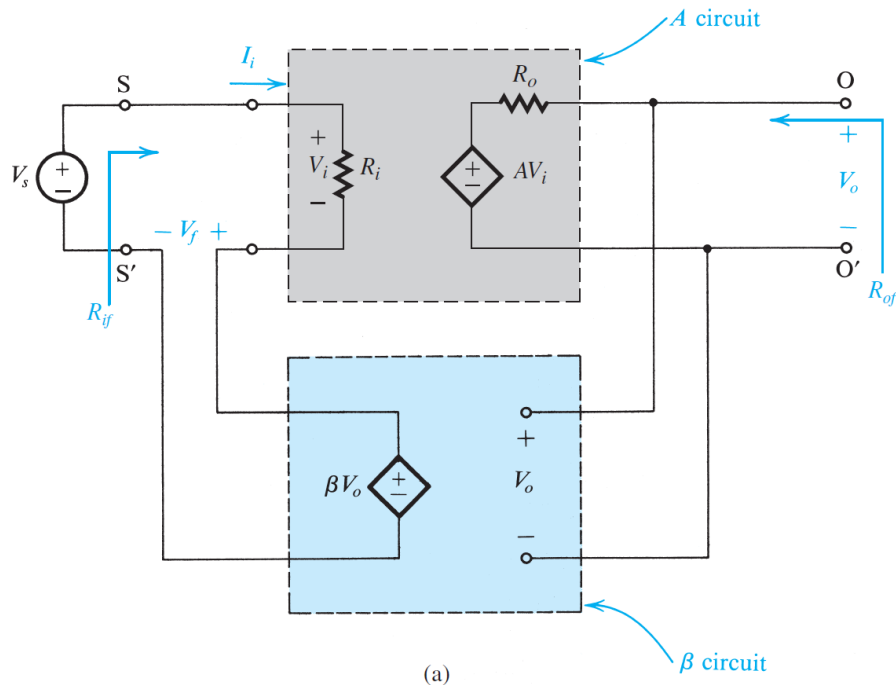


$$V_i = \frac{V_s}{1 + A\beta}$$

$$I_i = \frac{V_i}{R_i} = \frac{V_s}{(1 + A\beta)R_i}$$

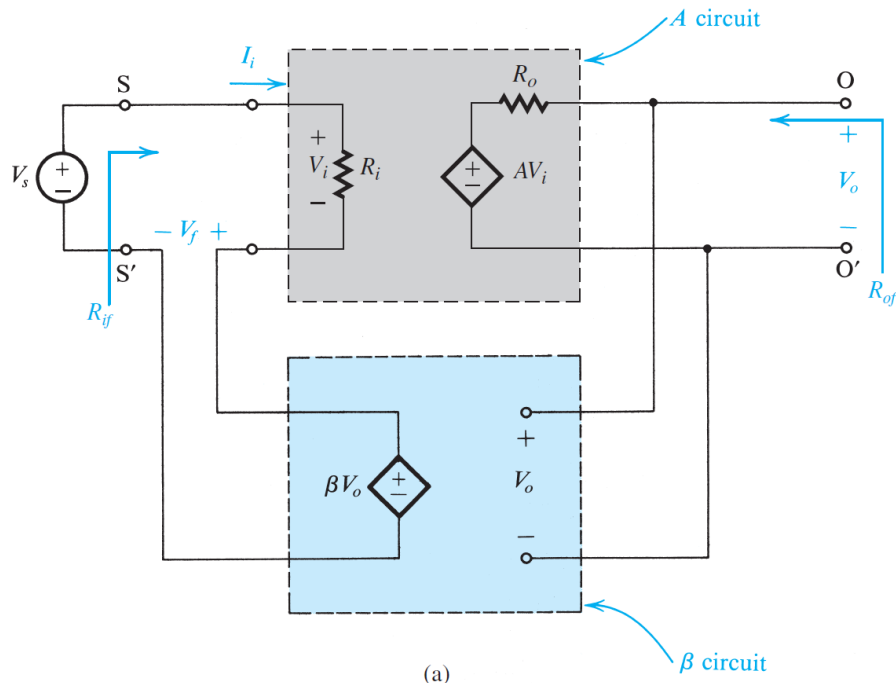
$$R_{if} \equiv \frac{V_s}{I_i}$$

$$R_{if} = (1 + A\beta)R_i$$



The series-shunt feedback amplifier: (a) ideal structure; (b) equivalent circuit.

Systematic Analysis of Feedback Voltage Amplifiers



$$R_{of} \equiv \frac{V_x}{I_x}$$

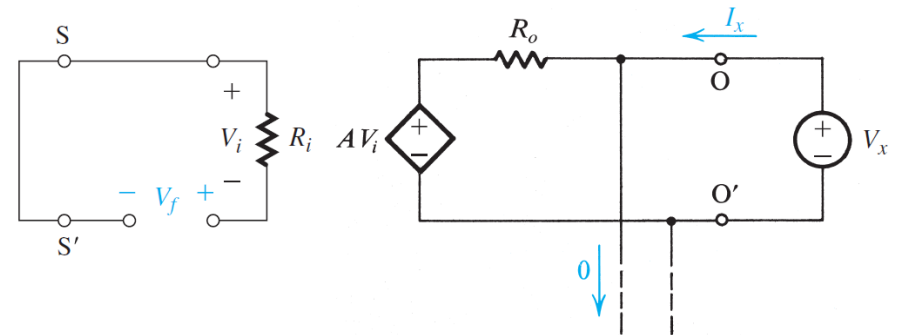
$$I_x = \frac{V_x - AV_i}{R_o}$$

$$V_i = -V_f$$

$$V_f = \beta V_o = \beta V_x$$

$$V_i = -\beta V_x \quad \Rightarrow \quad R_{of} = \frac{R_o}{1 + A\beta}$$

$$I_x = \frac{V_x(1 + A\beta)}{R_o}$$



Homework8-due June 10th 11p.m.



SEDRA/SMITH Book chapter 9 problem 9.85, 9.87,
chapter 11 problem 11.2, 11.11, 11.28, 11.37